

① Suppose the mass of electron decreased by 25%. How will it affect the Rydberg Constant?

i) it remains unchanged

ii) It becomes $\frac{1}{4}$ th.

iii) It reduces to 75% of original value.

iv) It is doubled.

② The charge on the electron and proton is reduced to half. Let the present value of Rydberg Constant is R . What will be the new value of Rydberg Constant.

a) $R/2$ b) $R/4$ c) $R/8$ d) $R/16$

③ For the same electronic transition in the following atom or ion, the frequency of the emitted radiation will be maximum for →

a) H-atom, b) D-Atom, c) He^+ ion d) Li^{2+} ion.

④ The orbit from which when electron will jump in other orbit energy may be absorbed but not emitted out will be → a) first orbit b) 2nd orbit, c) 7th orbit
d) infinite orbit.

5) Which of the following quantity for an electron revolving round the H-nucleus is independent to the mass of electron → a) Distance from nucleus b) K.E c) P.E,
d) Speed.

6) How much energy is needed for an electron revolving in 2nd orbit of He^+ ion, in order double its angular momentum → a) 40.8 eV b) 2.55 eV c) 10.2 eV
d) 12.09 eV.

⑦ The dissociation energy of H_2 is 430.53 kJ/mol . If H_2 is exposed to radiation energy of wavelength 253.7 nm , what % of radiant energy will be converted into kinetic energy $\rightarrow$ Ans $\rightarrow 8.68\%$.

⑧ O_2 undergoes photochemical dissociation into one normal oxygen atom and one oxygen atom 1.967 eV more energetic than normal. The dissociation of O_2 into two normal atoms of oxygen requires 498 kJ/mol . What is the minimum wavelength effective for photochemical dissociation of O_2 .
Ans $\rightarrow 1740.2 \text{ \AA}$

⑨ A photon of 300 nm is absorbed by a gas and then reemits two photons. one reemitted photon has wavelength 496 nm . Calculate energy of other photon reemitted out. $\rightarrow 2.625 \times 10^{-19} \text{ J}$

10) Calculate the wavelength of first line of Lyman series of 10 times ionised Sodium atom and compare with the wavelength of first line of Balmer series of H -atom.
Ans $\rightarrow 6565 \text{ \AA}$: times?